

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First Semester of B. Tech (CE/IT/EC/CSE) Examination****MAY 2018****CE141 / CE103 Computer Concepts & Programming****Date: 15.05.2018, Tuesday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I		
Q – 1	Do as directed.	
1.	State True or False: (1 mark for each correct true or false)	[04]
	i) C program execution begins from main() . ANS: True	
	ii) isupper(c) function checks whether c is lower case letter or not. ANS: False	
	iii) A “\n ” character instructs the computer to move the control to the next line. ANS: True	
	iv) C keywords can be used as variable names. ANS:False	
2.	The operator “++” is known as _____ operator. ANS: increment (1 mark for correct ans)	[01]
3.	Describe the use of scanf() in C. 1 mark for correct explanation	[01]
4.	What is software? 1 mark for Definition	[01]
Q – 2 a	Differentiate : Entry control loop and Exit control loop At least 2 differences must be written. 1 mark for each difference	[02]
Q – 2 b	Answer the following Questions. [Any Three]	[12]
i)	Draw and explain Basic Structure of C Program. 1 mark for diagram of basic structure of C program. 0.5 mark for each section explanation	
ii)	Introduce int, float, char and double data type in C. 1 mark for explanation of each data type. Neither example nor syntax is written in any of them then cut 1 mark. One line example is acceptable.	
iii)	Write down short note on goto statement in C. 1 mark for use of goto statement, 1 mark for syntax, 1 mark for writing types of goto and 1 mark for example.	
iv)	What is type conversion? Explain Implicit and Explicit type conversion.	

	1 mark for definition or use of type conversion, 1 mark each for explanation of implicit and explicit type conversion and 1 mark for Example. One line example is acceptable.	
v)	Compare: printf(), putchar(), fprintf() and puts(). 1 mark for each function explanation. Neither example nor syntax is written in any of them then cut 1 mark. One line example is acceptable.	
Q - 3 a	What is the use of simple if..else statement? Explain in detail with example. 1 mark for explanation of if..else statement, 1 mark for syntax, 2 mark for example.	[04]
	OR	
Q - 3 a	Give brief introduction of dynamic memory allocation function malloc() and calloc(). 2 mark for malloc() and 2 mark for calloc(). Neither example nor syntax is written in any of them then cut 1 mark. One line example is acceptable.	[04]
Q - 3 b	Attempt following Program.[Any Two]	[10]
i)	Write a program to find out given number is odd or even. 1 mark for declaration, 1 mark for scanning of variable, 2 marks for logic and 1 mark for displaying output.	
ii)	Write a program which will calculate addition, subtraction and multiplication of two number using switch-case statements. 1 mark for declaration, 1 mark for scanning of variable, 1 mark for switch case, 1 mark for logic and 1 mark for displaying output.	
iii)	Write a program which will print 1 to 10 number using loop. 1 mark for declaration, 1 mark for correct looping structure, 2 marks for logic and 1 mark for displaying output.	
iv)	Write a program to swap two numbers using pointers. 1 mark for pointer declaration, 1 mark for scanning of variable, 2 marks for logic and 1 mark for displaying output.	
SECTION – II		
Q - 4	Answer the question below.	
1.	State True or False: By default a function returns int data type value. ANS: True	[01]
2.	Match the following.	[04]
	i) sizeof() A) It clear console screen ii) clrscr() B) It gives size of structure. iii) NULL C) It indicates file termination. iv) EOF D) It is string termination character ANS: i) → B), ii) → A), iii) → D), iv) → C) 1 mark for one correct match	

3.	Describe String in C with suitable example. 1 mark for correct definition or explanation.	[01]
4.	What is Recursion? 1 mark for correct definition or explanation.	[01]
Q – 5 a	What is flow chart? Draw symbols of flowchart with its purpose. 1 mark for explanation of flowchart. 0.5 marks each for 1 symbol of flowchart with its meaning or name, so at least 6 symbols must be in answer.	[04]
	OR	
Q – 5 a	List down any four file handling functions. Write down use of fopen() function in detail. 2 marks for list down any for file handling functions, 1 mark for fopen() explanation and 1 mark for either syntax or program. One line example is acceptable.	[04]
Q – 5 b	Answer the following Questions. [Any Two]	[10]
i)	What is array? Explain in detail types of array with one example. 1 mark for definition of array, 3 marks for types of array (1 mark each for 1 array type) and 1 mark for Example	
ii)	Explain in detail any five categories of operator in C. 1 marks each for one category of operator. In answer must have at least 5 operators with its explanation, Only operator name is written then accordingly give 1 or 2 marks.	
iii)	Give name of different types of categories of function in C. Explain any two categories with example. 2 marks for list down categories of function, 1 mark each for explanation of 2 categories. 0.5 mark each for example or syntax of 2 categories.	
iv)	Write a short note on structure with example. 2 mark for explanation of Structure, 1 mark for syntax and 2 mark for example.	
Q – 6 a	Elaborate in details any four string handling functions of <string.h> header file. 1 mark for each function. Neither example nor syntax is written in any of them then cut 1 mark. One line example is acceptable.	[04]
	OR	
Q – 6 a	Define pointer. How to create and initialize pointer? Narrate any two advantages of pointer. 1 mark for pointer definition, 1 mark for pointer creation, 1 mark for pointer initialization and 0.5 marks for Advantages of pointer, minimum 2 advantages must be written.	[04]

Q – 6 b	Attempt following Program.[Any Two]	[10]
i)	Write a program to find out length of the string. 1 mark for declaration, 1 mark for scanning of variable, 2 marks for logic and 1 mark for displaying output. with or without string handling function both are allowed.	
ii)	Write a program to find out given number is positive, negative or zero. 1 mark for declaration, 1 mark for scanning of variable, 2 marks for logic and 1 mark for displaying output.	
iii)	Write a program which takes array of 5 integer number and find out minimum number from array. 1 mark for declaration, 1 mark for scanning of variable, 2 marks for logic and 1 mark for displaying output.	
iv)	Write a program to find out factorial of a given number using function. 1 mark for declaration, 1 mark for scanning of variable, 2 marks for logic and 1 mark for displaying output.	
